AMARA RAJA

Gotta be a better way

AMARON BRUTE DRAWING PORTAL

(Business Requirement Document)

AMARON BRUTE™

HI LIFE BATTERIES

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AMARA RAJA Gotta be a better way	Title **Amaron Brute Drawing Portal**	Ref. No : IAE/MPT/BRD/25-26 Rev. No : 00 Date : 16-12-25

1.Summary

In the Motive Power segment, battery tray and layout drawings play a critical role in ensuring accurate fitment across diverse vehicle models and applications. At present, these drawings are shared manually by the Application Engineering team upon request from Marketing.

This approach leads to delays in customer response, increased dependency on specific individuals, and inefficiencies in handling frequent and repetitive requests. Additionally, the absence of a centralized system raises the risk of version mismatches and inconsistent data usage.

The Amaron Brute Drawing portal provides a platform to access all required drawings and critical documents in one application. Marketing teams can directly search, download, and submit online requests for modifications or new drawings.

2.Project Summary

The Amaron Brute Drawing Portal is proposed as an intranet-based solution to centralize and streamline drawing management in the Motive Power business. It will host all layout and tray drawings, searchable by vehicle make, model, voltage, capacity, and rating, while supporting online drawing requests and updates. Currently more than 2000 drawings are available and the same will be uploaded in to drawing portal.

3. Objective

To digitalize and automate the drawing management process, providing Marketing teams with easy access to drawings and Application Engineering with a structured update process

4. Project Scope

- Categorize & Upload all drawings by vehicle make, model, Ah rating, capacity, voltage, and any special features.
- Provide a search functionality that enables Marketing teams to quickly locate the required drawings.
- Submit online requests for drawing modifications or new drawings that are automatically routed to the Application Engineering team with mail communication and portal notifications.
- Engineering team uploads and maps all developed drawings in the portal for new drawings and modification requests.
- Additional features of document library & training material to be maintained.
- There will be two level users will work on this portal , one as admin(Engg. Side) who will manage & control all data base & the other as user(MKTG) to download the available drawings & request for drawings.

5.Business Process Overview

5.1 Existing Process

- Users (MKTG) teams send request through E-mail to Engineering team for drawing requests.
- Engineering team will check if the required drawing is available in data base if its available, it will be forwarded, if its not available a new drawing will be developed and will be forwarded.
- With current enquiries we receive from MKTG, about 60% of drawings are already available in data base and this percentage will get increase as we develop more drawings.

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5.2 New Proposed Process

- Marketing team can search and download drawings directly from the portal, using the required parameters to locate the relevant drawings
- Drawing Modification requests or new drawings drawing requirements shall be submitted through the portal, ensuring standardized inputs and complete traceability
- Once the request is submitted in the portal, automated mail communication should come to the Application Engineering team, along with a portal notification.
- Application Engineering develops a new drawing or revises an existing drawing and uploads it in the portal against the request raised by Marketing user. After upload, automated mail communication should be sent to the intended Marketing user, along with a portal notification.
- The revision and version controls and record of the same should be available in portal for tracking.

6. Business Requirements

Sl. No	Requirement	Description
1	Master Data upload (Bulk)	Developer will upload all master data initially, and Engineering will prepare and share the master data.
3	Search & Download	Marketing can search drawings by vehicle model, make, AH capacity, voltage
4	Drawing Request in portal	Drawing request in portal for modifications and / or new drawings
5	Mail communications	Mail notifications and portal notification for new drawings requests or drawing modifications.
6	Drawings Control	Maintain drawing history with revision date
7	Technical Data upload	Upload picture, technical documents, training materials & videos. Mktg can view & download
8	Master Data management	Admin should have controls to modify/adda/delete the mater data partially and fully for required fields as applicable.

7.Access Control & User Roles

7.1 Admin -Engg. Side

Holds full control of the portal, managing master data, uploading, modifying, and deleting documents, manuals, and training materials, monitoring portal activities, and providing access to new users as required.

7.2 Users -MKTG team

Access to search, view and download drawings using relevant input parameters and submit online requests for drawing modifications or new drawing development

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Annexure 1: Amaron Brute Drawing Portal Detailing

1. Overview

Amaron Brute Drawings Portal should contain all drawing data in a centralized database. Authorized users should be able to search the database using selected filters to view and download drawings, submit modification requests, and create new drawing requests. The portal should also provide access to all Amaron Brute related documents and images stored in the database.

2. Home Page

2.1 Amaron Brute Drawing portal Home Page should contain the following two primary options, displayed with related images/icons.

- Drawing Search
- Search by Drawing Number

These two options act as the main access points for finding drawings from the database.

2.2 Navigation Menu

Home Page navigation bar of the Amaron Brute Drawing portal contains the following modules:

- Amaron Brute Drawing (Drawing search / Search by Drawing number)
- Drawing Request
- Drawing Modification
- Documents Download

3. Drawing Search

- Users can search drawings from the database by entering/selecting the drop-down fields and view or download the drawings.
- All database initially will be provided by engineering team for bulk upload, however this database will keep increase continuously as we expand our business in this segment, the system/IT logics created for upload of database to be flexible enough to facilitate the new additions or modification or deletion in the database by engineering team who should have control as admin to make all these changes.

3.1 Drawing Search Process flow

The drawing search process is step by step from the list of drop down fields,

- Selecting the Drawing Search option on the Home Page opens the Drawing Search page
- Search page contains drop-down and input fields for the user to select or enter all required parameters.
- Users can search drawings from the database by entering or selecting the drop-down fields and view or download the drawings. However, if an exact drawing is not available, drawings with closest dimensions should be suggested based on the specified dimensional tolerances.

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- Non-standard drawings should also be suggested. The suggested non-standard drawings should include all available Ah ratings based on the specified voltage and dimensions.

Note: System/IT logics created for:

- Prompt logic created to be used for entering dimensions in the specified fields, with numbers not less than 3 and not more than 4 digits.
- Upload of drawings in the database to be flexible enough to facilitate modification or replacement for OEM-wise voltage and Ah ratings.
- If an exact match drawing is not available, the system should suggest non-standard drawings completely based on user-entered dimensions within tolerance limits of ±20 mm (length), ±10 mm (width), and ±20 mm (height).

3.1.1 User Request for Modification

- User can modify the selected drawing from the database by raising a modification request.
- Modification Request option available on Drawing Search page to raise new request for selected drawing. Comment box available for user to enter the modification requirements.
- User can modify the selected drawing from the database by raising a modification request.

3.1.2 Responder Upload (Application Engineering)

- Responder modifies the drawing as per the request and uploads and maps the approved drawing accordingly in the portal by entering and selecting dropdown fields.
- Once uploaded, automated mail communication goes to the requesting user, along with portal notification, and the drawing becomes immediately available for view/download in the Drawing Search page.

Note : System/IT logic to be created to systematically store the modified drawing into the database once it is uploaded.

4. Search by Drawing Number

- Users can search drawings from the available database by entering the drawing number, and view or download the drawing.

Workflow

- Search by Drawing Number option is available on the Home Page for the user to search drawings from the database using the drawing number.
- Search page contains an input field for the user to enter the drawing number, the drawing number prompt has logic to be enter specified Numeric Alpha characters as defined for drawing number same prompt to be given if any user entering wrong formats for drawing number search.
- User can enter the drawing number and search from the available database, and view or download the required drawing

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Note : System/IT logics created for:

- System/IT logics created to accept only the defined format of 8 digits and 1 alphabet, arranged as two digits, space, three digits, space, three digits, space, and one alphabet (example: 25 600 001 A).
- Drawing Search page also contains an option for the user to raise a modification request using the drawing number for the selected drawing

5. New Drawing Request.

- User can raise a request for a new drawing by entering and selecting the dropdown fields based on the requirements.

5.1 User Request Process:

- This page contains the New Drawing Request form for user to request a new drawing by entering and selecting the dropdown fields.
- Form contains all required fields in a single page, enabling the user to enter details most of the filed to be drop down selection as covers in drawing search, the additional fields included to be entered in the specified fields.
- Once the request is submitted, automated mail communication should goes to the Application Engineering Team.

5.2 Admin Upload (Application Engineering)

- Responder develops the new drawing as per the request and uploads and maps the approved drawing accordingly in the portal by entering and selecting dropdown fields.
- Once uploaded, automated mail communication goes to the requesting user, along with a portal notification, and the drawing becomes immediately available for view/download.

Note: System/IT logic to be created to systematically store the new drawing into the database once it is uploaded.

6. Document Management

- This page contains the Documents option on the Home Page. Users can search, view, and download the required documents and pictures from the database..
- Documents Page contains all the motive power documents and images in the database. Users can search from the database and view or download the required documents and pictures.
- Documents Page contains all documents and pictures in the database in the form of Presentations (PPT), Manuals (PDF), Videos (MP4), JPEG, PNG, and other formats.

Note : system/IT logics created for upload of documents to be flexible enough to facilitate the new additions or modification or deletion in the data base by engineering team who should have control as admin to make all these changes.

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7. User Roles and Access Rights

7.1 Admin Engineering Team)

- Admin has full access to the portal to facilitate new additions, modifications, or deletions, and can manage user roles and permissions.

7.2 Users (Marketing Team / Field Team)

- Users have access to search, view, and download drawings, documents, and pictures, and can submit requests for drawing modifications or new drawing development.

Note: System/IT logic created to be flexible for giving access to new users by the Engineering team, who hold admin control to provide access.

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Annexure 2: Screen Functionality Reference

1.Home Page

Home Page of the Amaron Brute Drawing Tracker consists of a left-side navigation menu and main screens.

Left-side navigation menu includes the following screens:

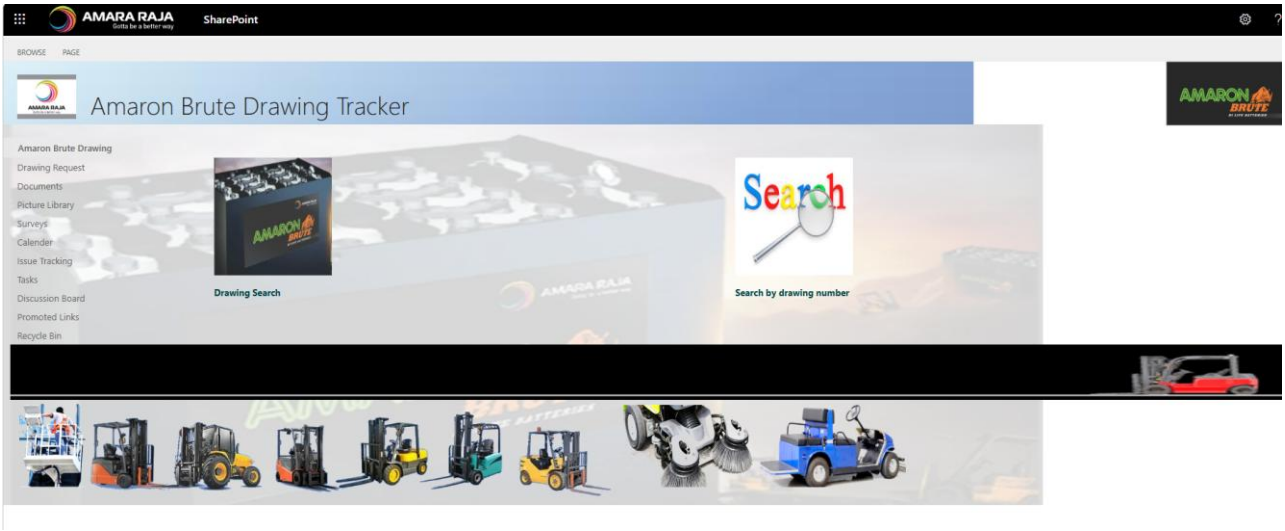
- Amaron Brute Drawing Screen (Both Admin & User)
- Drawing Request Screen (Both Admin & User)
- Drawing Modification Request Screen (Both Admin & User)
- Documents Library Screen (Both Admin & User)
- Master Database Screen (For Admin only)

1.1 Amaron Brute Drawing Screen

This is the default screen when the portal is opened and provides two primary options for drawing access.

- **Drawing Search** – Used to search drawings based on vehicle parameters and battery parameters
- **Search by Drawing Number** – Used to directly access drawings using the drawing number if availble.

Note : These options are the main access points for finding and downloading drawings, and on clicking each option, respective screen will open



1.2Drawing Search Screen:

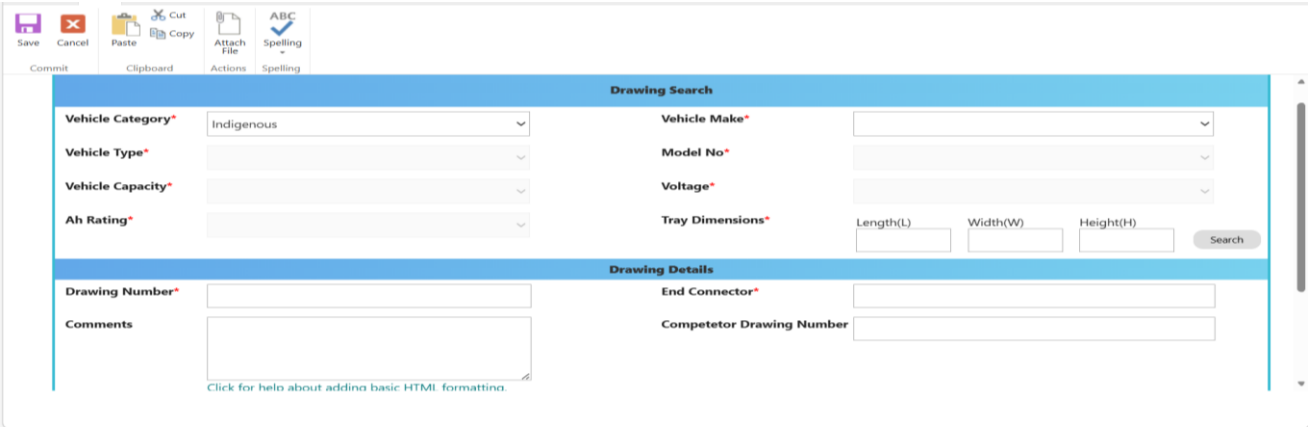
The Drawing Search page consists of two screens, one for admin to upload and manage drawings, and another for users to view and download drawings.

- Admin Screen
- User Screen

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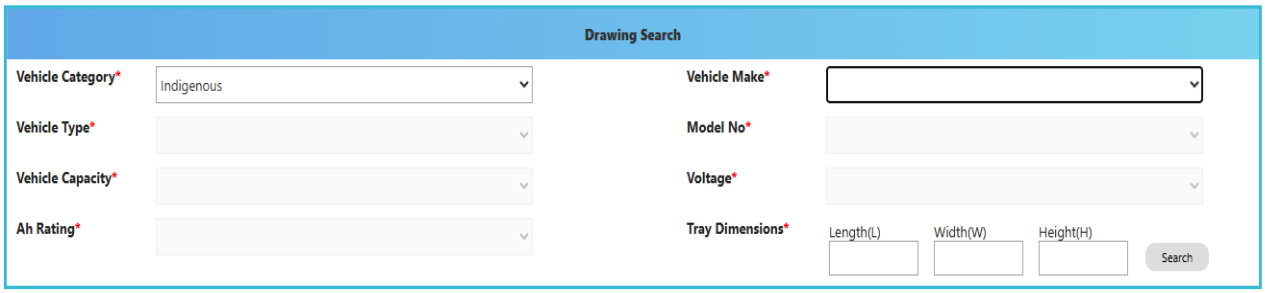
1.3 Admin Screen: (For New Requirements)

- Admin screen should have the option to upload new drawings by entering and selecting the required fields.
- Screen should have the option to upload drawing files using the attach file option and a save option to store the drawing and details in the database.



1.4 User Screen

- Search page should contain drop-down and input fields for selecting and entering all required parameters.
- Users can search drawings from the database by selecting values from drop-down fields and entering inputs in the specified fields, and view and download the drawings.
- System should provide prompt logic for dimension input fields, allowing only numeric values between 3 and 4 digits.

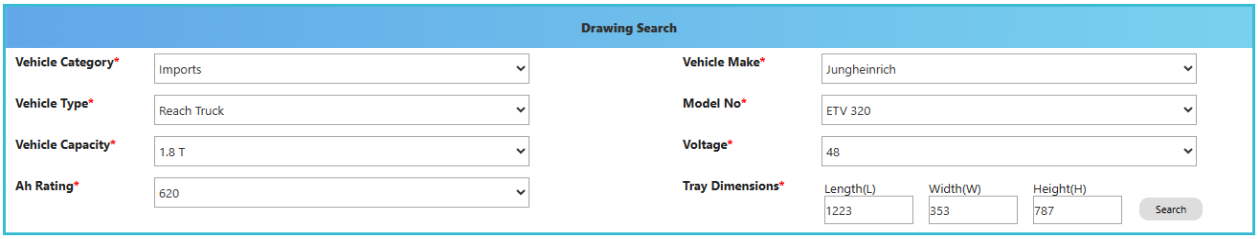


1.5 Drawing Suggestion Logic (Standard Drawing)

- System should display a standard drawing only when all user inputs, including Ah rating and dimensions, match the available drawing data

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1.6 Input



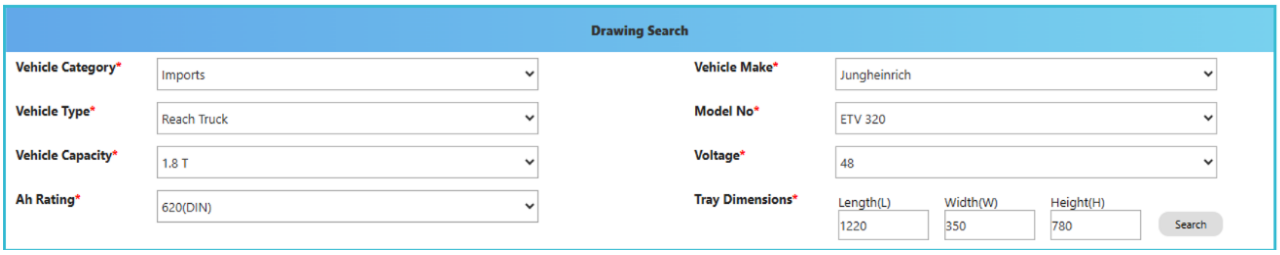
1.7 Output (Standard Drawings)

ID	Drawing Number	AH Rating	Voltage	Connector	Length	Width	Height	Comments
411	18 502 553 A	560	48	SR 350A	1223	353	787	null

1.8 Drawing Suggestion Logic (Non-Standard Drawing)

- If an exact match drawing is not available, the system should suggest non-standard drawings completely based on user-entered dimensions within tolerance limits of ±20 mm (length), ±10 mm (width), and ±20 mm (height).
- **Note Point (Non-Standard Drawings):** It should be mentioned that suggested non-standard drawings are based on closest dimension tolerances. Users should select the drawing that best matches their requirements.

1.9 Input



1.10 Output

Non Standards:

ID	Drawing Number	AH Rating	Voltage	Connector	Length	Width	Height	Comments
411	18 502 553 A	560	48	SR 350A	1223	353	787	null
430	18 502 279 A	620	48	SR 350A	1220	350	785	null
484	17 500 614 A	620(DIN)	48	SR 350A	1220	350	784	null
500	17 500 695 A	620(DIN)	48	SR 350A	1220	352	784	null
520	17 500 694 A	620(DIN)	48	SR 350A	1220	352	784	null
541	17 500 965 A	620(DIN)	48	SR 350A	1210	342	770	null

2.Search by drawing number

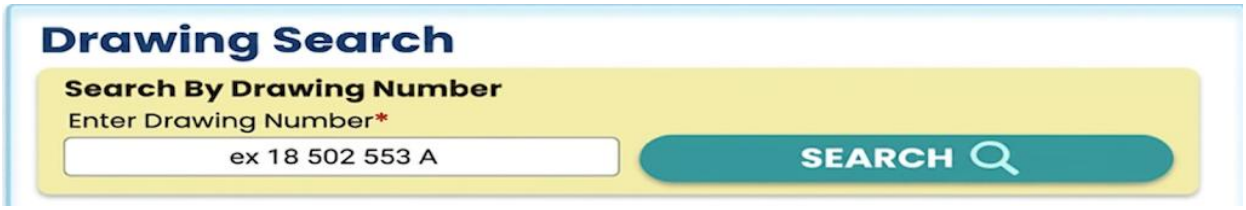
2.1User Screen

- Users should be able to search drawings by entering a Drawing Number in the input field as per the defined standard format.

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- System shall accept only the defined format of 8 digits and 1 alphabet, arranged as two digits, space, three digits, space, three digits, space, one alphabet (e.g., 18 502 553 A) and validate the input before processing the search.

2.2 Input



2.3 OutPut

ID	Drawing Number	AH Rating	Voltage	Connector	Length	Width	Height	Comments
411	18 502 553 A	560	48	SR 350A	1223	353	787	null

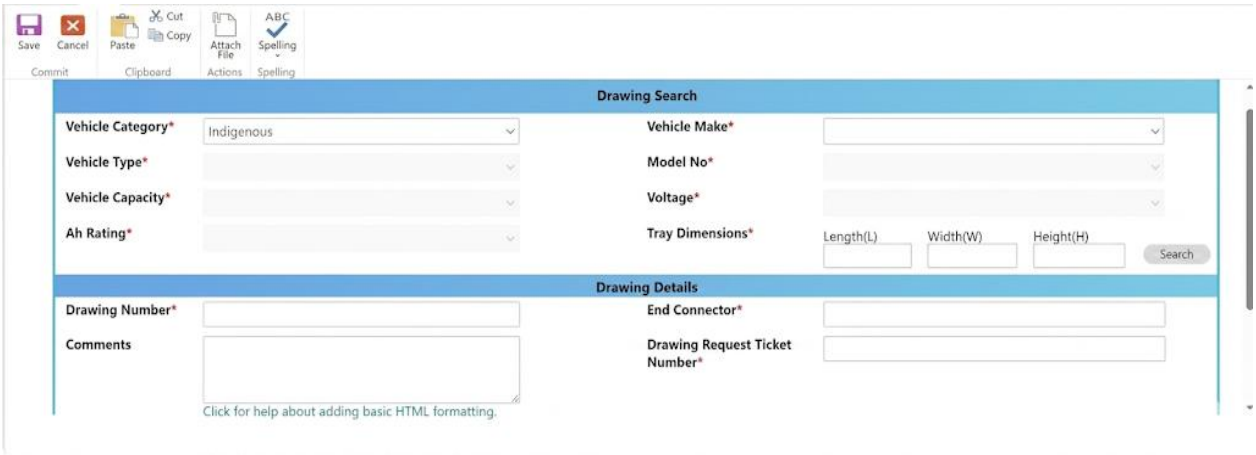
3.Drawing Request

Drawing Request page consists of two screens, one for admin to manage and respond to requests, and another for users to raise new drawing requests, with automatic ticket generation and closure based on the request.

- Admin Screen
- User Screen

3.1Admin Screen

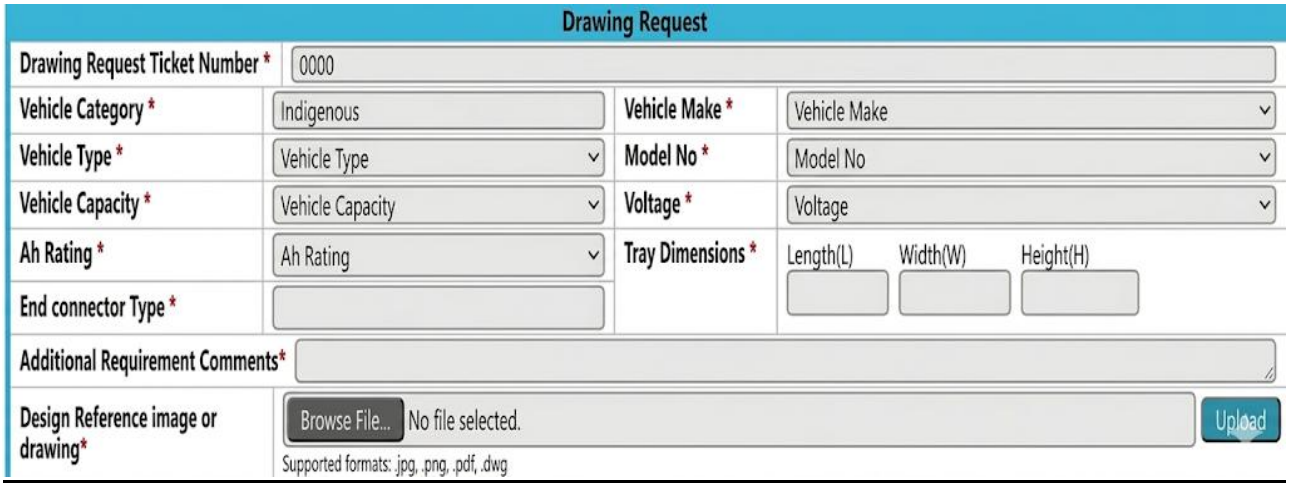
- Admin can review all drawing requests and upload drawings against the request.
- Admin can upload new drawings against the request, and the ticket should close automatically, and mail communication should go to the user once the drawing is uploaded.
- Once a new drawing is uploaded, it should be saved in the database.



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3.2User Screen

- User can raise a new drawing request by entering and selecting required details, adding comments, and uploading reference.
- User can submit the request, and the system should automatically generate a ticket, and automated mail communication should go to the Engineering team.



4.Drawing Modification Request

Drawing Modification Request page consists of two screens, one for admin to manage and respond to modification requests, and another for users to raise drawing modification requests, with automatic ticket generation and closure based on the request.

- Admin Screen
- User Screen

4.1 Admin Screen

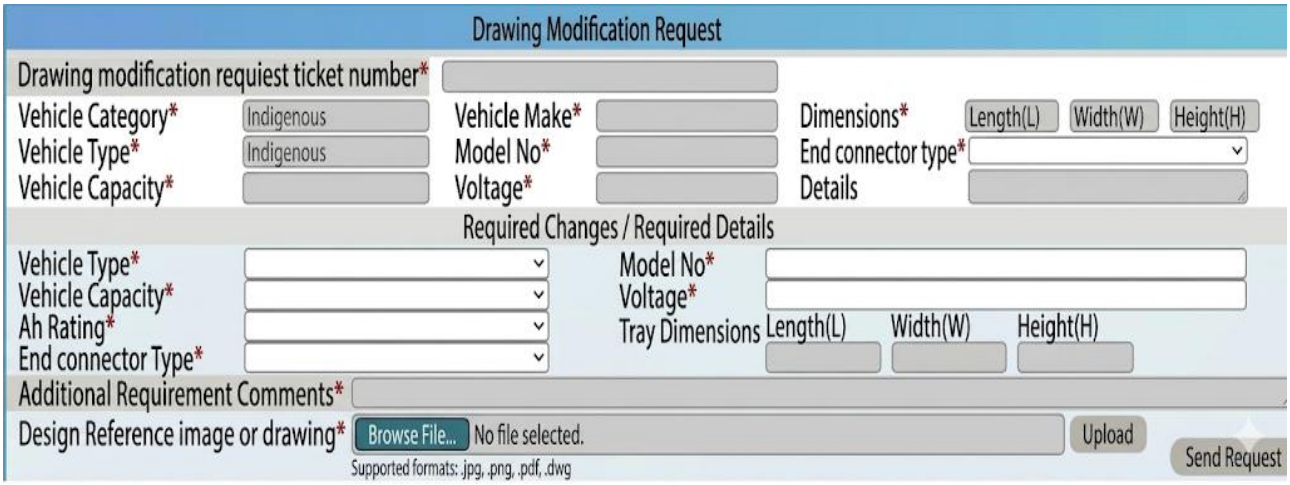
- Admin can review all drawing modification requests and upload modified drawings against the request.
- Admin can upload modified drawings against the request, and the ticket should close automatically, and mail communication should go to the user once the drawing is uploaded.
- Once the modified drawing is uploaded, it should be saved in the database as a revision of the drawing.



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4.2User Screen

- User can raise a drawing modification request by entering the Drawing Number and selecting required details, adding comments, and uploading reference images/drawings.
- User can submit the request, and the system should automatically generate a ticket, and automated mail communication should go to the Engineering team.



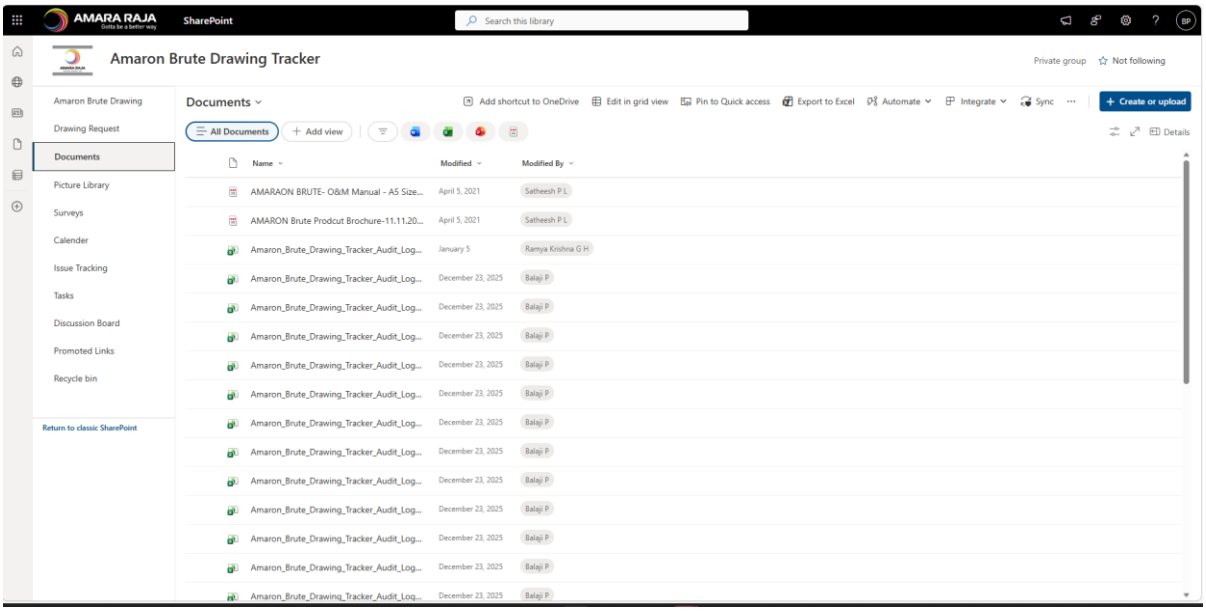
5.Documents

Document Search page is designed with two different screens to support different user roles

- Admin Screen
- User Screen

5.1Admin Screen

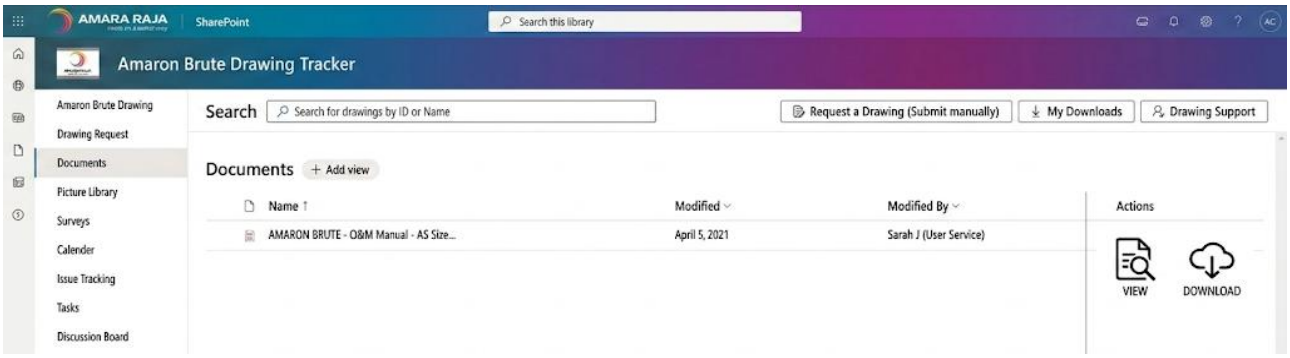
- Admin/Responder can Create or upload and manage documents in the system, including edit, modify, and delete actions.



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5.2User Screen

- User can search documents using available fields and can view and download them.



6.Master Database Screen

- To facilitate the data as upto as per changing business needs, the all master data may get changes with time with new MHEs or New OEMs or change in requirements.
- Master Database Screen should display all database records, and Admin/ should have control to add, modify, and delete the data and drawing partially or fully, once its modified or deleted the changes should reflect into user screen as well.(user notification may not required)
- Master data will be prepared and shared by the Engineering team, and developer will upload all master data initially. All new and modified drawings against requests will be managed and uploaded by the Engineering team.
- Master data should be displayed in a structured tabular format for Admin, with options to view, upload, modify, and delete records.
- In Master Data, Admin should have full control to perform partial or complete modifications, delete existing drawings, and upload new drawings, with all changes reflected in the system.

6.2 Master Data Screen Reference (Admin)

Make	Vehicle Type	Model	Capacity	Voltage (V)	Ah	Drawing No	Length (mm)	Width (mm)	Height (mm)	Socket Type	Action	
Linde	Reach	E16H	1.6 (t)	48	775	21 600 549	1220	420	780	DIN 350A	Modify	Delete
Linde	Forklift	EFG 218	1.8 T	48	750	23 600 345	830	735	630	DIN 160A	Modify	Delete
Toyota	Forklift	EFG 320	2 T	48	625	19 500 258	825	625	625	DIN 320A	Modify	Delete
Toyota	Forklift	8FB30	3/3.5T	80	460	24 600 345	948	815	589	SBX 350A	Modify	Delete
Godrej	Stacker	ES 1530	1.5 T	24	225	25 600 162	750	180	570	SR 175A	Modify	Delete

---End---

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